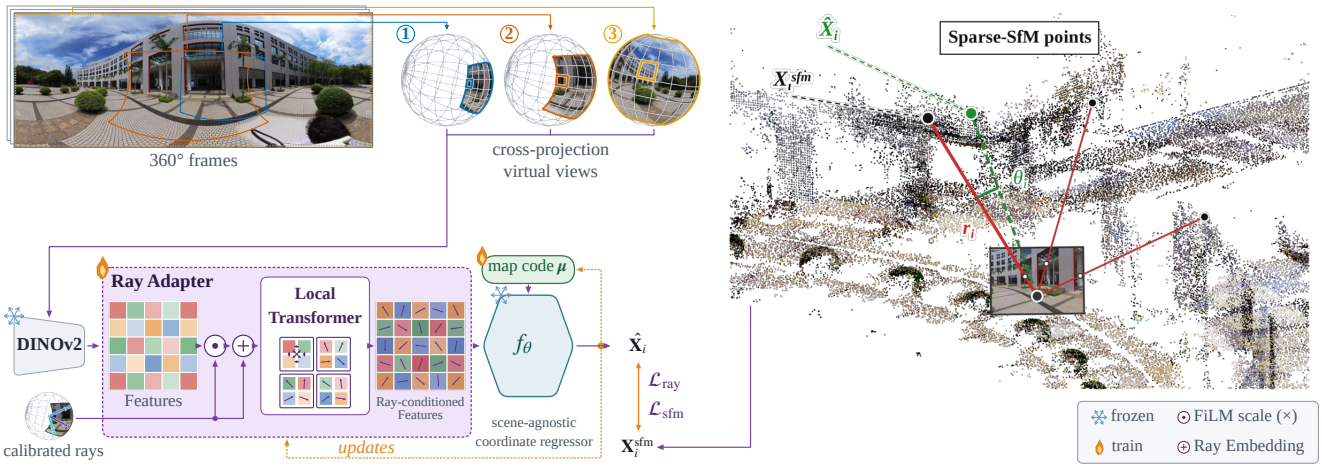

360° frames
①
②
③
cross-projection
virtual views
Sparse-SfM points
$\hat{X}_i$
X_i^{sfm}
θ
r_i
DINOv2
Ray Adapter
Features
Local
Transformer
Ray-conditioned
Features
map code μ
f_θ
scene-agnostic
coordinate regressor
$\hat{\mathbf{X}}_i$
$\mathcal{L}_{\text{ray}}$
$\mathcal{L}_{\text{sfm}}$
$\mathbf{X}_i^{\text{sfm}}$
calibrated rays
updates
frozen
train
FiLM scale (×)
Ray Embedding